		Potential at R, $V = \ldots = -2.7 \times 10^{-13}$ J Hence, work done = electric potential energy = $eV = 4.3 \times 10^{-13}$ J
20	**D**	Option A: False. Charge from X to Y decreases distance from Z, so potential increases. Option B: False. Charges are further apart, so potential energy decreases. Option C: False. Charges are further apart, so force between them decreases. Option D: True. By process of elimination. [Proof]$E_W = \frac{kQ}{r^2}$ (downwards), $E_X = \frac{kQ}{(\sqrt{2}r)^2} = \frac{kQ}{2r^2}$ (away from Z along diagonal) $E_{net} = \sqrt{E_W^2 + E_X^2 - 2E_W E_X \, cos \, cos \, 135°} = \sqrt{\left(\frac{kQ}{r^2}\right)^2 + \frac{1}{4}\left(\frac{kQ}{r^2}\right)^2 - 2\left(\frac{kQ}{r^2}\right)\left(\frac{kQ}{2r^2}\right)\left(-\frac{1}{\sqrt{2}}\right)} = 1.399\frac{kQ}{r^2}$ When charge moves from X to Y, $E_Y = \frac{kQ}{r^2}$ (right) $E'_{net} = \sqrt{E_W^2 + E_Y^2} = \sqrt{\left(\frac{kQ}{r^2}\right)^2 + \left(\frac{kQ}{r^2}\right)^2} = 1.414\frac{kQ}{r^2}$
21	**C**	$\frac{V_S}{V_P} = \frac{N_S}{N_P} \Rightarrow r.m.s.\ V_S = \frac{1600}{200} \times 25 = 200\,V$